

LOKESHPRASANTH GADESULA

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Senior Software & Applied AI Engineer | Distributed Systems • Backend Infrastructure • LLM Evaluation & Agentic Workflows | Python, Go, Spark, Kafka, AWS

U.S. Work Authorized (OPT / STEM OPT) | Senior Software & Applied AI Engineer

Executive Summary:

Senior Backend and Applied AI Engineer specializing in the intersection of high-throughput distributed systems, large-scale data engineering, and LLM evaluation architectures. With over 6 years of experience across cloud-native environments, I engineer the production data pipelines, evaluation harnesses, and resilient backend systems that power modern AI workloads.

Core Technical Pillars:

- **Applied AI & LLM Systems:** Designing automated evaluation benchmarks, hallucination detection pipelines, synthetic data generation, and tool-augmented agentic workflows for enterprise LLM applications.
- **Data & Streaming Infrastructure:** Architecting high-throughput batch and streaming pipelines using Spark, Kafka, Airflow, and AWS to process operational and analytical datasets at scale.
- **Backend & Platform Engineering:** Building resilient, low-latency microservices and asynchronous API architectures (Python, Go, C++, SQL) backed by cloud-native infrastructure and deep observability.

Master Technical Skills Matrix:

Discipline	Technologies & Frameworks
Languages	Python (FastAPI, Asyncio, PyTest), C++, Go, JavaScript / TypeScript, SQL (PostgreSQL, Redshift, T-SQL)
Data & Streaming	Apache Spark (PySpark), Apache Kafka, Apache Airflow, AWS Glue, Amazon Redshift, Athena, PostgreSQL, Redis
AI, LLMs & Evaluation	LLM Benchmarking, Evaluation Frameworks (Ragas, DeepEval, MLflow), Agentic Workflows, Tool-Use Validation, RAG Architecture, LangChain, LlamaIndex, DSPy, Synthetic Data Generation
Cloud & Distributed Infra	AWS (S3, EMR, ECS, Lambda, RDS, CloudWatch), Google Cloud Platform (GCP), Docker Containerization, Kubernetes, Terraform (IaC)
DevSecOps & Observability	GitHub Actions (CI/CD), OpenTelemetry, Prometheus, Grafana, Linux Kernel/POSIX Internals, Secure SDLC, Zero-Trust Principles

Work Experience:

Pioneer Solutions | Suwanee, GA, USA

Senior Software Engineer & AI Evaluation Specialist | Jul 2025 – Present

- Architected end-to-end evaluation and benchmarking frameworks to validate LLM-generated code, reducing functional regression rates by 40% across complex asynchronous and cloud-native test suites.
- Engineered distributed testing harnesses using **Python and AWS ECS** to evaluate reasoning, API tool calling, and deterministic outputs for agentic systems handling thousands of test scenarios daily.
- Designed low-latency backend APIs and automated validation microservices in **Python (FastAPI, Asyncio) and Go**, reducing integration test execution times by 35%.
- Implemented distributed telemetry and structured logging pipelines across containerized microservices using **OpenTelemetry, Prometheus, and Grafana** to monitor model inference anomalies and system bottlenecks.
- Enforced schema validation contracts with Pydantic and automated data quality checks, eliminating malformed outputs in downstream data stores.

Cybersoft Technologies | Houston, TX, USA

Full Stack & Distributed Systems Engineer | Sep 2023 – Jun 2025

- Built event-driven ingestion pipelines utilizing **Apache Kafka and Apache Spark on AWS EMR**, processing millions of events daily with sub-second delivery to analytics datastores.
- Developed scalable backend REST/gRPC endpoints in Python backed by **PostgreSQL and Redshift**, improving multi-service query throughput by 30%.
- Automated distributed ETL workflows and dependency directed acyclic graphs (DAGs) using **Apache Airflow**, eliminating pipeline drift and reducing SLA breaches to under 1%.
- Built operational monitoring dashboards with **React, TypeScript**, and telemetry platforms, giving platform teams real-time visibility into streaming pipeline health and worker node resource consumption.
- Containerized microservices with Docker and standardized GitHub Actions CI/CD workflows, reducing deployment cycle times across staging and production.

Graduate Academic & Systems Research | Southeast Missouri State University

Graduate Systems Researcher & Developer | Aug 2023 – May 2025

- Researched distributed computing architectures, LLM retrieval pipelines, and algorithmic efficiency across multi-node Linux environments.
- Formulated empirical benchmarking suites in Python, Go, and C++ to evaluate memory bandwidth saturation, inter-process communication latency, and cache efficiency under heavy concurrent workloads.
- Authored open-source benchmark repositories with automated test harnesses, reproducible environment configurations, and documentation on GitHub.

Zognu | Hyderabad, India

Frontend & Distributed Systems Engineer | Sep 2021 – Jul 2023

- Implemented data ingestion interfaces and high-performance administrative dashboards using **React, TypeScript, and Python** microservices.
- Integrated distributed **Spark-based** aggregation pipelines to serve near-real-time analytics to operational interfaces.
- Provisioned resilient AWS cloud infrastructure using **Terraform**, cutting deployment turnaround by 25% through standardized infrastructure-as-code (IaC).
- Optimized relational database queries and indexing strategies in PostgreSQL, eliminating transactional locks under peak usage.

NullClass | Bengaluru, India

- Built foundational Python backend APIs, database indexing schemas, and authentication layers for platform services.
- Automated serverless job orchestration using **AWS Lambda and CloudWatch**, minimizing compute idle time by 20%.
- Established CI/CD deployment automation using **GitHub Actions and Docker**, reducing build failure rates and standardizing developer staging environments.

Featured Scale & Architecture Projects:

Enterprise LLM Evaluation & Red-Teaming Harness

- *Repository:* github.com/lokeshgadesula
- *Tech Stack:* Python, Docker, AWS Lambda, PostgreSQL, PyTest, LangChain, LlamaIndex, Ragas
- Built an automated benchmarking pipeline that stress-tests LLM tool calling, structured JSON output adherence, and edge-case code generation across 1,000+ synthetic engineering scenarios.
- Designed modular metric evaluators measuring semantic drift, context precision, and hallucination rates, integrating regression check gates into continuous integration workflows.

Scalable Event-Driven Ingestion & Analytics Engine

- *Repository:* github.com/lokeshgadesula
- *Tech Stack:* Python, Apache Kafka, Apache Spark, Apache Airflow, AWS S3, Amazon Redshift
- Designed a dual-stream batch and real-time processing engine that ingests high-velocity streaming data via Kafka, executes deduplication and stateful aggregation in Spark, and materializes analytical tables into Redshift.
- Orchestrated multi-stage DAGs in Airflow with automated backfill handling, SLA monitoring, and data lineage tracking.

Education:

- **Master of Science in Applied Computer Science** – Southeast Missouri State University (*Graduated May 2025*)
- **Bachelor of Technology in Computer Science** – Kakatiya University (*Graduated June 2021*)